Executive summary

Problem and decision. A Colombian services group collects its invoices about 22 days later than agreed, which ties up working capital. Two collections analysts can give personal follow-up to only 20 customers per day, out of about 135 with overdue invoices, and today they choose by balance and due date. The decision we support is which 20 customers to contact each day.

Data. We use the company's own anonymized ERP records, used under written authorization: 19,262 balances and 40,519 movements. The target, an invoice paid more than 30 days after its due date, is not in the data, so we built it, measuring delay up to the last cash receipt. This leaves 15,822 labeled invoices, 32.84% of them late.

Method. We trained models on invoices issued from January 2022 to June 2025, using 11 predictors known at issue, and evaluated them forward in time. Logistic regression matched a random forest, gradient boosting and a neural network within validation noise, so we kept it as the simplest to explain. The test period, July 2025 to September 2026, was queried only once. A daily list ranks customers by late-payment probability times balance.

Main result. On unseen data the model reaches a precision-recall AUC of 0.744, against 0.386 for the base rate. With a quota of 20 customers, its list captures 47.4% of the money still at risk that ended up late, versus 37.9% for the current rule, and it wins on 254 of 296 business days.

Main limit. The backtest does not show that contacting customers shortens delays. Probabilities run low: where the model predicts 32%, 52% were late. It also misses more late invoices in the group's smaller companies (61% versus 32%, in validation, on only 57 late invoices).

Recommendation. Adopt with conditions. Use the list to order work, not to promise savings. Retrain on recent history now, because late invoices reached 38.6% against a 31–34% reference. Monitor four metrics quarterly. Validate the benefit with a randomized pilot by customer before reporting any effect.